BharatKart Sales & Business Performance Analysis Report

1. Introduction

This project presents an end-to-end data analytics workflow on BharatKart sales data. The dataset was initially provided in raw CSV format and was processed using Python for data cleaning. SQL was then used to perform analysis and extract key business insights. Finally, an interactive dashboard was developed to visualize performance metrics and trends.

The goal of this project is to convert raw transactional data into actionable insights that help in understanding business performance and improving decision-making.

2. Objectives

- To clean and pre-process raw sales data using Python
- To analyze structured data using SQL queries
- To build an interactive dashboard for visualization
- To identify trends in sales, customer behavior, and operations
- To generate insights for business improvement

3. Dataset Description

The dataset contains sales transaction records of BharatKart.

Key Fields:

- Order Date
- Customer ID
- City
- Product Category
- Product Name
- Order Value
- Payment Mode
- Discount
- Delivery Days
- Customer Rating

The cleaned dataset was enriched with time-based columns such as Month and Year for better trend analysis.

4. Tools and Technologies Used

- Python (Pandas) – Data cleaning and preprocessing
- SQL – Data analysis and querying
- Power BI – Dashboard creation and visualization
- CSV/Excel – Data storage

5. Data Cleaning Process (Python)

The raw dataset contained inconsistencies and required preprocessing.

Steps Performed:

- Removed duplicate records
- Handled missing/null values
- Standardized categorical columns (City, Category)
- Converted order date into datetime format
- Extracted Month and Year for analysis
- Ensured consistency in numeric fields like order value, discount, and rating

Challenges Faced

- The raw dataset contained missing and inconsistent values, especially in columns like City and Category
- Some records had incomplete or incorrect entries, which required cleaning before analysis
- Standardizing categorical data (e.g., different city name formats) was challenging
- Ensuring data accuracy before performing SQL analysis required additional validation steps

This ensured the dataset was clean, structured, and analysis-ready.

6. Data Analysis Using SQL

SQL queries were used to generate business insights.

Key Analysis Performed:

- Total Revenue calculation

- Total Orders count
- Average Delivery Time
- Average Customer Rating
- Repeat Customer identification
- City-wise order distribution
- Category-wise revenue contribution
- Product-level revenue analysis
- Payment mode distribution
- Monthly sales trend analysis

7. Dashboard Overview

The dashboard titled “BharatKart Sales & Business Performance Dashboard” provides a comprehensive view of business performance.

Key KPIs (Top Cards):

- Total Revenue: ₹1.11 Billion
- Total Orders: 18K
- Average Delivery Time: 7.50 days
- Average Rating: 3.01
- Repeat Customers: 5K

8. Visual Analysis & Insights

8.1 Revenue by Product Category

Revenue is almost evenly distributed across categories:

- Electronics (~25.37%)
- Home (~25.27%)
- Grocery (~24.54%)
- Fashion (~24.82%)

☞ Insight: Business is well-diversified with no single category dominating revenue.

8.2 Orders by City

Top cities by order volume:

- Hyderabad
- Chennai

- Mumbai
- Pune
- Bangalore
- Delhi

☞ Insight: Metro cities contribute the highest number of orders, indicating strong urban demand.

8.3 Revenue by Product

Top-performing products:

- Shirt
- Sofa
- Mobile
- Rice
- Laptop

☞ Insight: Both essential goods (Rice) and high-value items (Laptop, Mobile) drive revenue.

8.4 Monthly Sales Trend

Sales fluctuate across months but show consistent performance overall

Peaks observed around mid-year and festive periods

☞ Insight: Seasonal trends impact sales; business can leverage this for targeted campaigns.

8.5 Payment Mode Analysis

Payment distribution:

- UPI (Highest)
- COD
- Card

☞ Insight: Digital payments (especially UPI) dominate customer preference.

8.6 Delivery & Customer Satisfaction

- Average delivery time: 7.5 days
- Average rating: 3.01

☞ Insight:

- Delivery speed needs improvement
- Moderate ratings indicate scope for better customer experience

8.7 Repeat Customers

- Around 5K repeat customers

☞ Insight: Customer retention is strong but can be further improved through loyalty programs

9. Key Business Insights (Summary)

- Revenue is evenly distributed across categories
- Metro cities are major contributors to sales
- UPI is the most preferred payment method
- Delivery time impacts customer satisfaction
- Seasonal trends influence sales performance
- Mix of essential and premium products drives revenue
- Repeat customers contribute significantly to business

10. Conclusion

This project successfully demonstrates a complete data analytics lifecycle:

- Data cleaning using Python
- Data querying and analysis using SQL
- Dashboard creation using Power BI

The dashboard provides clear insights into sales performance, customer behavior, and operational efficiency. These insights can help businesses improve decision-making, optimize delivery, and enhance customer experience.